

Week 5: CI/CD Pipeline with Azure DevOps

Objective

Build a CI/CD pipeline in Azure DevOps to run expense analysis automatically on a weekly or monthly schedule.

Capstone Tasks

- Build a pipeline that runs expense analysis weekly or monthly
- Output summary report as CSV
- Log or print a savings alert if expenses exceed a threshold

Tools Used

- Azure DevOps
- YAML Pipeline
- Python

Project Files

expenses_analysis.py

```
import pandas as pd
```

```
# Load expense data
```

```
df = pd.read_csv('expenses.csv')
```

```
# Monthly summary
```

```
monthly = df.groupby('category')['amount'].sum().reset_index()
```

```
monthly.to_csv('monthly_analysis.csv', index=False)
```

```
print('Monthly analysis saved successfully!')
```

```
# Savings alert
```

```
threshold = 5000

if df['amount'].sum() > threshold:

    print('ALERT: Expenses exceeded threshold!')

else:

    print('Expenses are within budget.')
```

expenses.csv

category,amount

Food,500

Travel,300

Shopping,700

Food,250

Travel,400

Azure DevOps YAML Pipeline

Create a file named azure-pipelines.yml

trigger:

- main

schedules:

- cron: "0 0 * * 0"

displayName: Weekly Expense Analysis

branches:

include:

- main

always: true

pool:

vmImage: 'ubuntu-latest'

steps:

- task: UsePythonVersion@0

inputs:

versionSpec: '3.x'

- script: |

pip install pandas

python expenses_analysis.py

displayName: 'Run Expense Analysis'

- task: PublishBuildArtifacts@1

inputs:

pathToPublish: 'monthly_analysis.csv'

artifactName: 'MonthlyReport'

Pipeline Steps

1. Trigger pipeline from the main branch.
2. Schedule automatic weekly execution.
3. Install Python dependencies.
4. Run expense analysis script.
5. Generate monthly_analysis.csv.
6. Print savings alert if threshold is exceeded.
7. Publish the CSV as a pipeline artifact.

sample Output

Console Log

Monthly analysis saved successfully!

ALERT: Expenses exceeded threshold!

Output File

monthly_analysis.csv containing category-wise monthly totals.

Deliverables

- Azure DevOps YAML pipeline
- Output file of monthly analysis